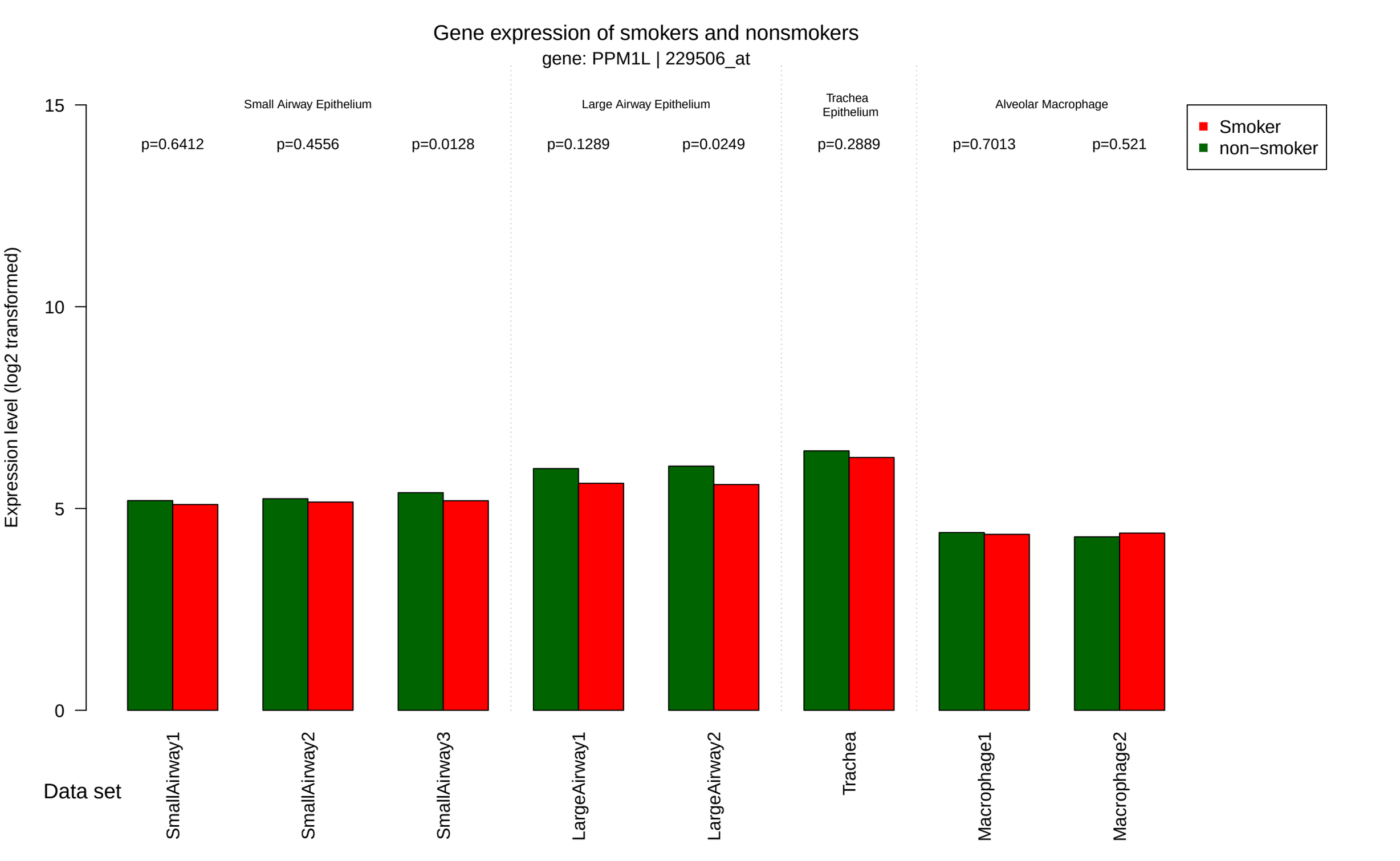
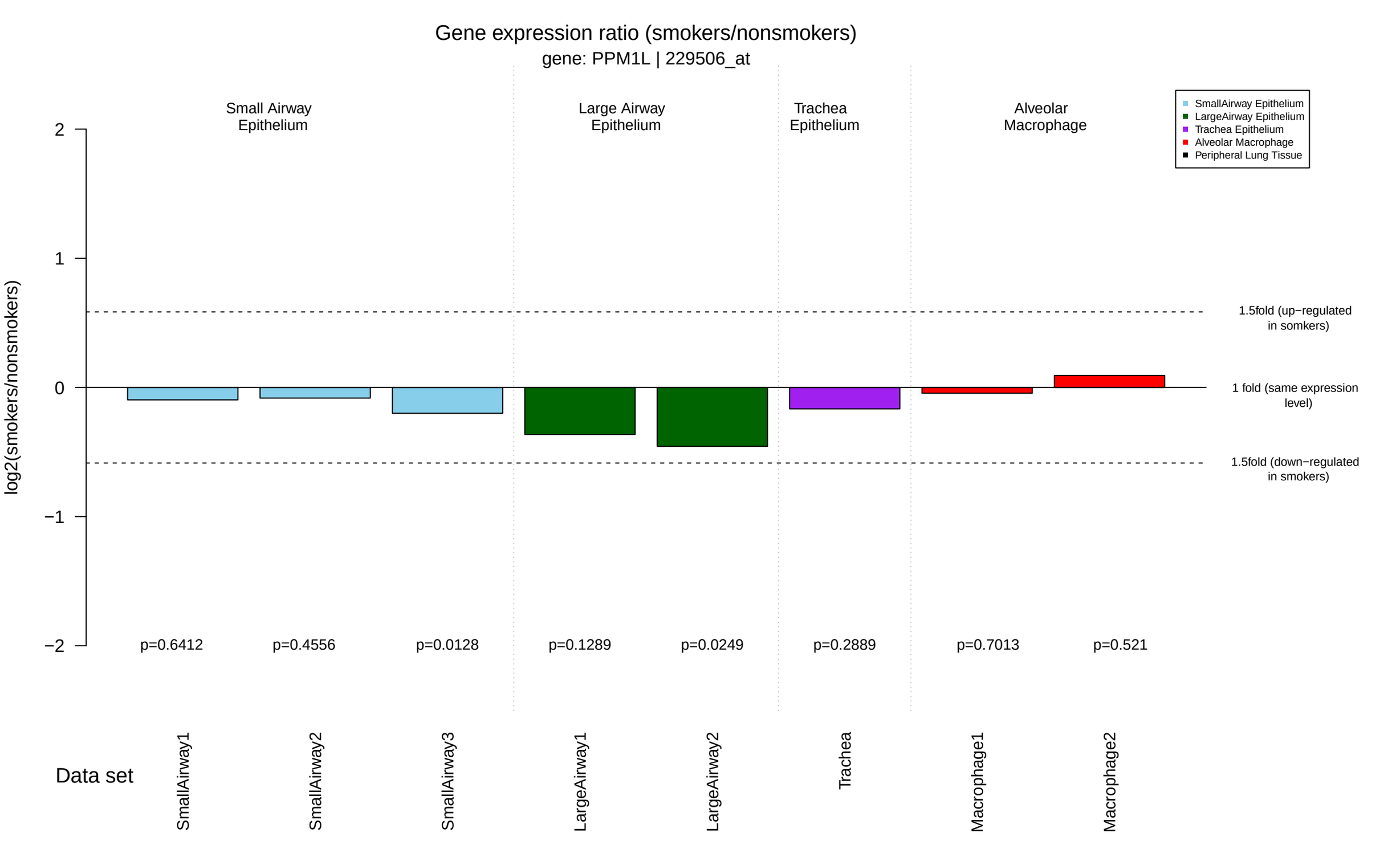
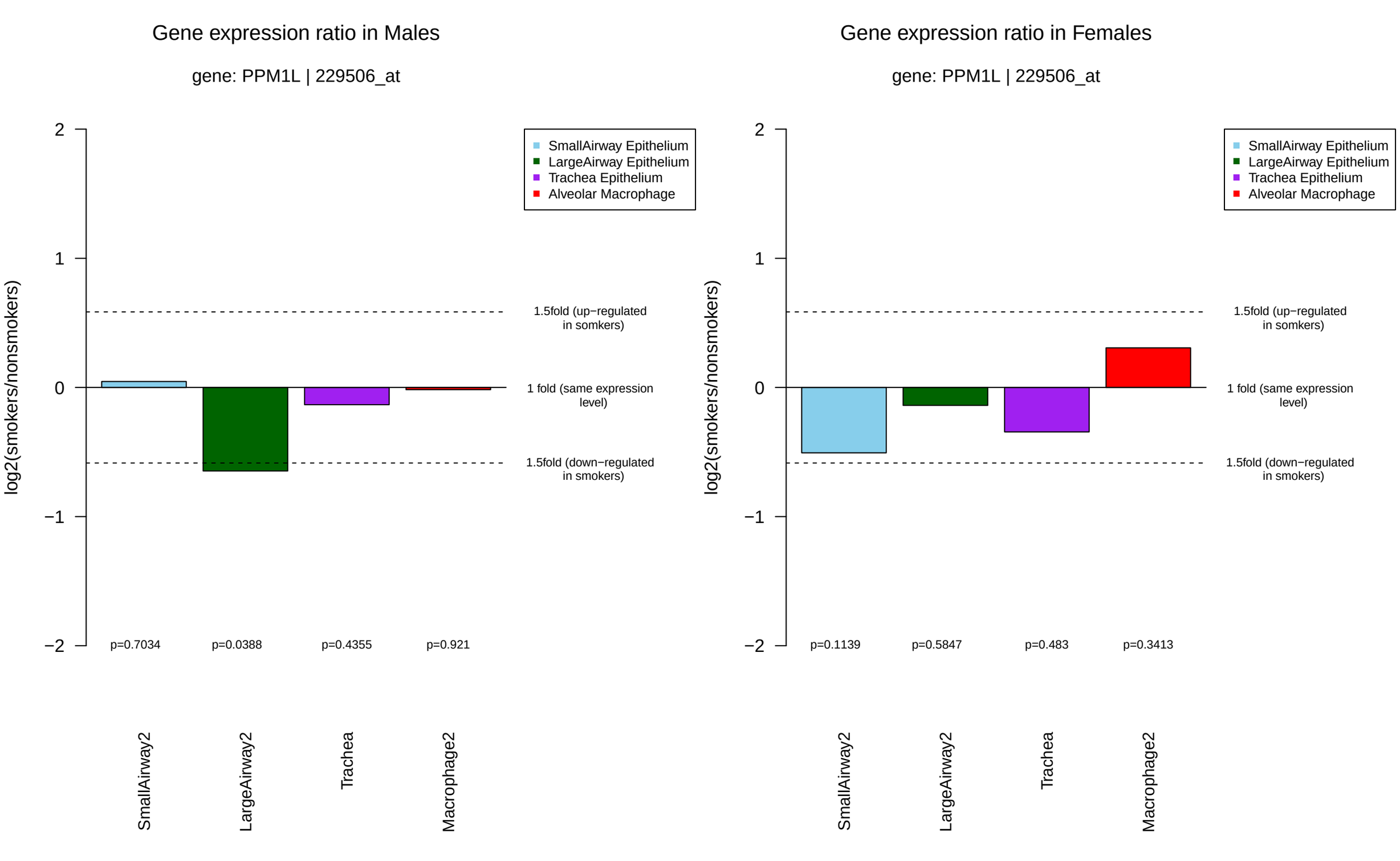
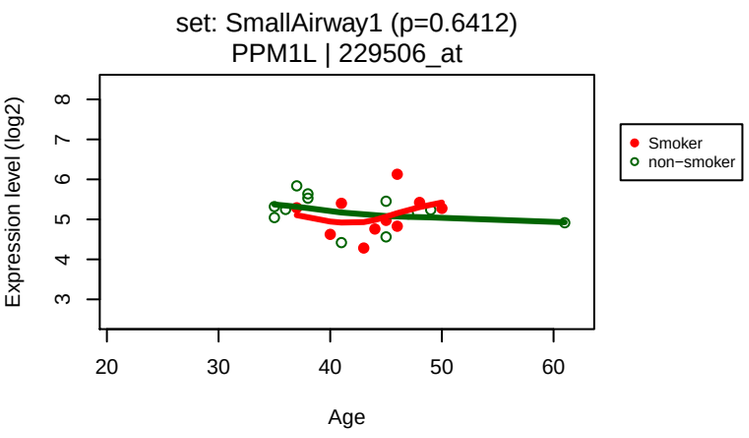
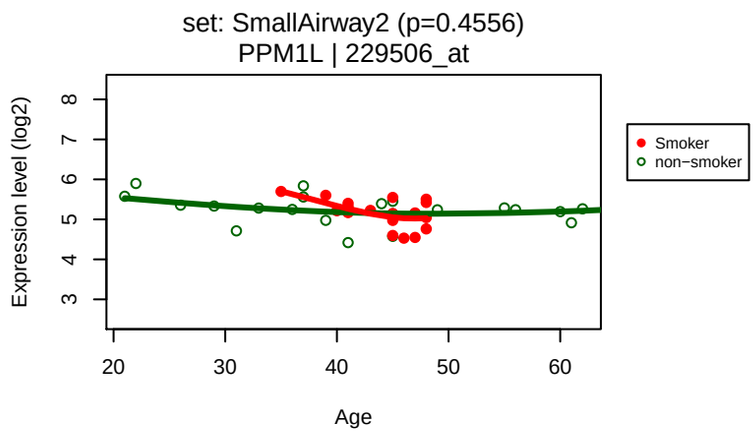
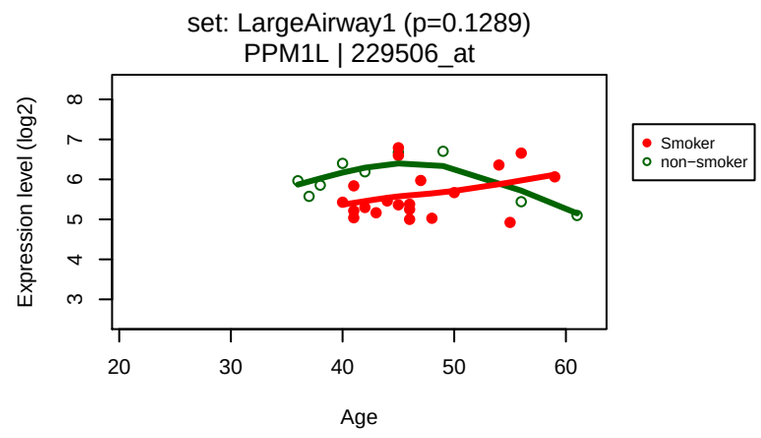

Gene expression of smokers and nonsmokers
gene: PPM1L | 229506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6412
p=0.4556
p=0.0128
p=0.1289
p=0.0249
p=0.2889
p=0.7013
p=0.521
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

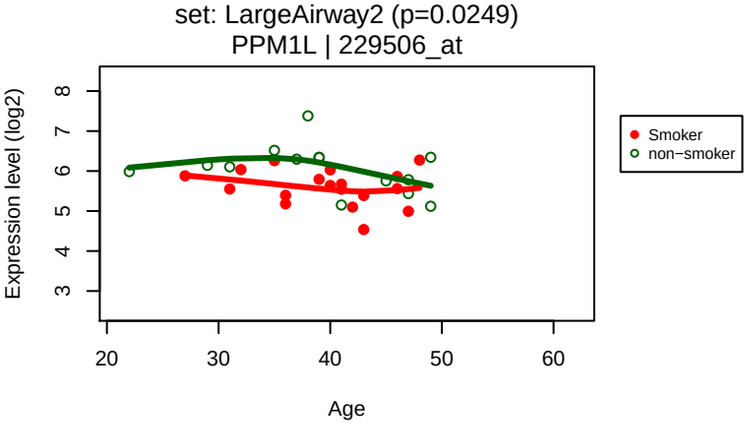

Gene expression ratio (smokers/nonsmokers)
gene: PPM1L | 229506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6412
p=0.4556
p=0.0128
p=0.1289
p=0.0249
p=0.2889
p=0.7013
p=0.521
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

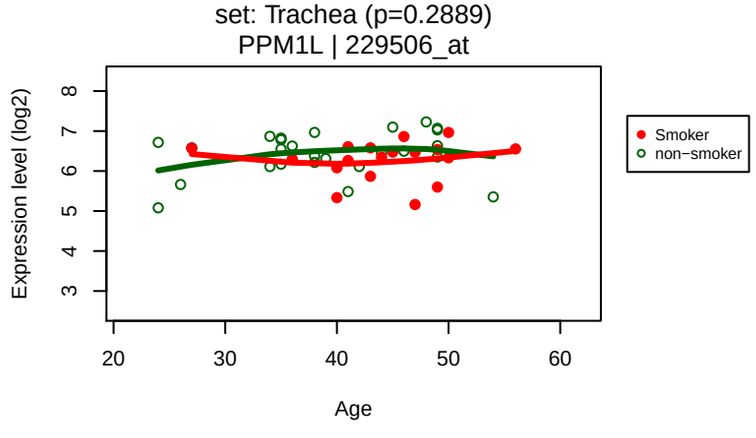

Gene expression ratio in Males
gene: PPM1L | 229506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7034
p=0.0388
p=0.4355
p=0.921
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1L | 229506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1139
p=0.5847
p=0.483
p=0.3413
SmallAirway2
LargeAirway2
Trachea
Macrophage2

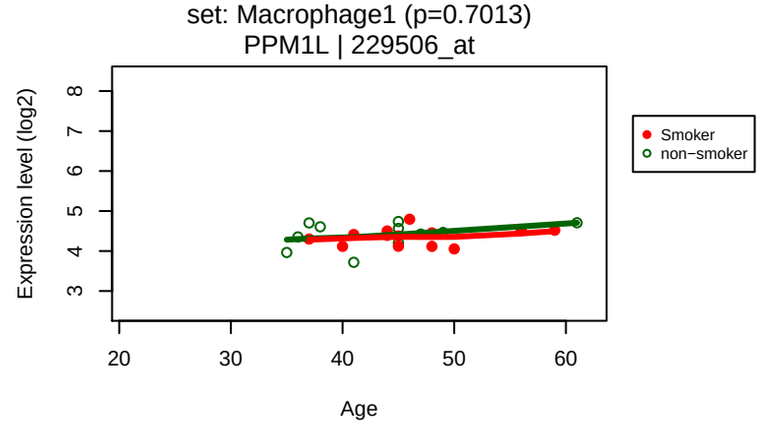

set: SmallAirway1 (p=0.6412)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker

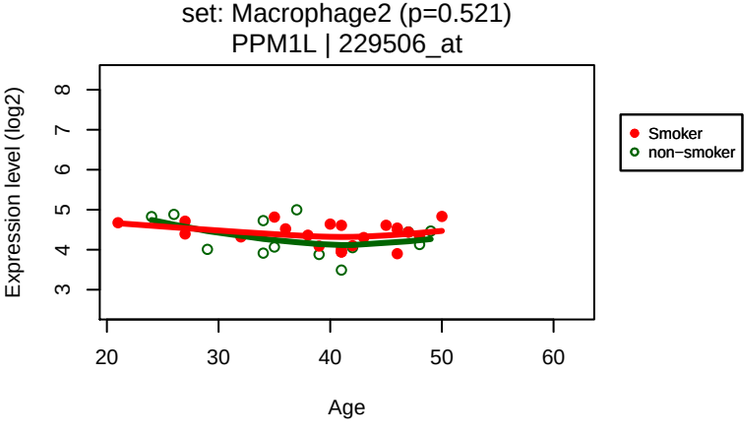

set: SmallAirway2 (p=0.4556)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker

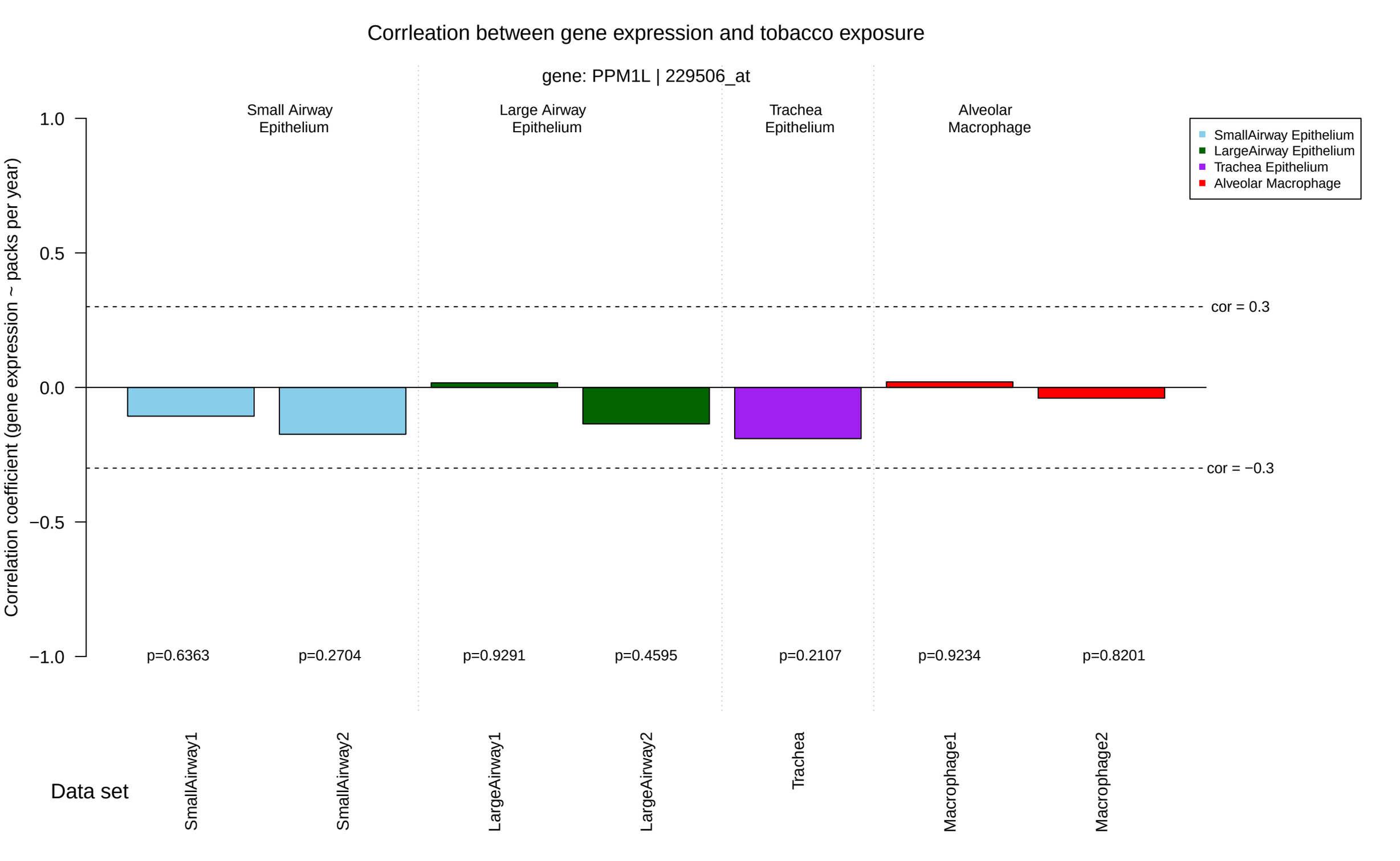

set: LargeAirway1 (p=0.1289)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0249)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2889)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7013)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.521)
PPM1L | 229506_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1L | 229506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6363
p=0.2704
p=0.9291
p=0.4595
p=0.2107
p=0.9234
p=0.8201
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2